Laboratorio de Microeconomía I

Centro de Investigación y Docencia Económicas
Maestría en Economía - 2025
Laboratorista: Arturo López
Tarea 8

Exercises

Ex. 1.60 Suppose $x_1(\mathbf{p}, y)$ and $x_2(\mathbf{p}, y)$ have equal income elasticity at $(\mathbf{p}^0, y^0)$. Show that

$$\frac{\partial x_1}{\partial p_2} = \frac{\partial x_2}{\partial p_1} \text{ at } (\mathbf{p}^0, y^0)$$

Ex. 1.63 The substitution matrix of a utility-maximising consumer's demand system at prices $(8, p)$ is

$$\begin{pmatrix} a & b \\ 2 & -1/2 \end{pmatrix}.$$

Find a, b, and p.

Ex. 3.G.3$^{\mathbf{B}}$ Consider the (linear expenditure system) utility function given in Exercise 3.D.6.

(a) Derive the Hicksian demand and expenditure functions. Check the properties listed in Propositions 3.E.2 and 3.E.3.

(b) Show that the derivatives of the expenditure function are the Hicksian demand function you derived in (a).

(c) Verify that the Slutsky equation holds.

(d) Verify that the own-substitution terms are negative and that compensated cross-price effects are symmetric.

(e) Show that $S(p, w)$ is negative semidefinite and has rank 2.

Ex. 3.G.9$^{\mathbf{C}}$ Compute the Slutsky matrix from the indirect utility function.

Appendix

Ex. 3.D.6$^{\mathrm{B}}$ (Mas Colell et al., 1995). Consider the three-good setting in which the consumer has utility function

$$u(x) = (x_1 - b_1)^{\alpha}(x_2 - b_2)^{\beta}(x_3 - b_3)^{\gamma}.$$

(a) Why can you assume that $\alpha + \beta + \gamma = 1$ without loss of generality? Do so for the rest of the problem.

(b) Write down the first-order conditions for the UMP, and derive the consumer's Walrasian demand and indirect utility functions. This system of demands is known as the *linear expenditure system* and is due to Stone (1954).

(c) Verify that these demand functions satisfy the properties listed in Propositions 3.D.2 and 3.D.3.

Proposition 3.G.3: The Slutsky Equation (Mas Colell et al., 1995).

Suppose that $u(\cdot)$ is a continuous utility function representing a locally nonsatiated and strictly convex preference relation $\succsim$ defined on the consumption set $X = \mathbb{R}^L_+$. Then for all (p, w), and $u = v(p, w)$, we have

$$\frac{\partial h_\ell(p,u)}{\partial p_k} = \frac{\partial x_\ell(p,w)}{\partial p_k} + \frac{\partial x_\ell(p,w)}{\partial w} x_k(p,w) \quad \text{for all } \ell, k.$$

or equivalently, in matrix notation,

$$D_p h(p,u) = D_p x(p,w) + D_w x(p,w) x(p,w)^\top.$$

Proposition 3.G.3 implies that the matrix of price derivatives $D_p h(p,u)$ of the Hicksian demand function is equal to the matrix

$$S(p,w) = \begin{bmatrix} s_{11}(p,w) & \cdots & s_{1L}(p,w) \\ \vdots & \ddots & \vdots \\ s_{L1}(p,w) & \cdots & s_{LL}(p,w) \end{bmatrix},$$

with

$$s_{\ell k}(p,w) = \frac{\partial x_\ell(p,w)}{\partial p_k} + \frac{\partial x_\ell(p,w)}{\partial w} x_k(p,w).$$

This matrix is known as the *Slutsky substitution matrix.* Note, in particular, that $S(p,w)$ is directly computable from knowledge of the (observable) Walrasian demand function $x(p,w)$.

Because $S(p,w) = D_p h(p,u)$, Proposition 3.G.2 implies that when demand is generated from preference maximization, $S(p,w)$ must possess the following three properties: it must be *negative semidefinite*, *symmetric*, and satisfy $S(p,w)p = 0$.

Theorem 1.11 The Slutsky Equation (Jehle & Reny, 2011).

Let $x(\mathbf{p}, y)$ be the consumer's Marshallian demand system. Let u^* be the level of utility the consumer achieves at prices $\mathbf{p}$ and income y. Then,

$$\underbrace{\frac{\partial x_i(\mathbf{p}, y)}{\partial p_j}}_{\text{TE}} = \underbrace{\frac{\partial x_i^h(\mathbf{p}, u^*)}{\partial p_j}}_{\text{SE}} \underbrace{- x_j(\mathbf{p}, y)\frac{\partial x_i(\mathbf{p}, y)}{\partial y}}_{\text{IE}}, \qquad i, j = 1, \ldots, n.$$

Theorem 1.15 Negative Semidefinite Substitution Matrix (Jehle & Reny, 2011).

Let $x^h(\mathbf{p}, u)$ be the consumer's system of Hicksian demands, and let

$$\sigma(\mathbf{p}, u) \equiv \begin{pmatrix} \frac{\partial x_1^h(\mathbf{p},u)}{\partial p_1} & \cdots & \frac{\partial x_1^h(\mathbf{p},u)}{\partial p_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial x_n^h(\mathbf{p},u)}{\partial p_1} & \cdots & \frac{\partial x_n^h(\mathbf{p},u)}{\partial p_n} \end{pmatrix},$$

called the *substitution matrix*, containing all the Hicksian substitution terms. Then the matrix $\sigma(\mathbf{p}, u)$ is negative semidefinite.

Theorem 1.16 Symmetric and Negative Semidefinite Slutsky Matrix (Jehle & Reny, 2011).

Let $x(\mathbf{p}, y)$ be the consumer's Marshallian demand system. Define the ijth Slutsky term as

$$\frac{\partial x_i(\mathbf{p}, y)}{\partial p_j} + x_j(\mathbf{p}, y)\frac{\partial x_i(\mathbf{p}, y)}{\partial y},$$

and form the entire $n \times n$ **Slutsky matrix** of price and income responses as follows:

$$s(\mathbf{p}, y) = \begin{pmatrix} \frac{\partial x_1(\mathbf{p},y)}{\partial p_1} + x_1(\mathbf{p}, y)\frac{\partial x_1(\mathbf{p},y)}{\partial y} & \cdots & \frac{\partial x_1(\mathbf{p},y)}{\partial p_n} + x_n(\mathbf{p}, y)\frac{\partial x_1(\mathbf{p},y)}{\partial y} \\ \vdots & \ddots & \vdots \\ \frac{\partial x_n(\mathbf{p},y)}{\partial p_1} + x_1(\mathbf{p}, y)\frac{\partial x_n(\mathbf{p},y)}{\partial y} & \cdots & \frac{\partial x_n(\mathbf{p},y)}{\partial p_n} + x_n(\mathbf{p}, y)\frac{\partial x_n(\mathbf{p},y)}{\partial y} \end{pmatrix}.$$

Then $s(\mathbf{p}, y)$ is symmetric and negative semidefinite.

Proposition 3.E.2 (Mas Colell et al., 1995):

Suppose that $u(\cdot)$ is a continuous utility function representing a locally nonsatiated preference relation $\succsim$ defined on the consumption set $X = \mathbb{R}^L_+$.

The expenditure function $e(p, u)$ is

(i) Homogeneous of degree one in p.

(ii) Strictly increasing in u and nondecreasing in p_ℓ for any ℓ.

(iii) Concave in p.

(iv) Continuous in p and u.

Proposition 3.E.3 (Mas Colell et al., 1995):

Suppose that $u(\cdot)$ is a continuous utility function representing a locally nonsatiated preference relation $\succsim$ defined on the consumption set $X = \mathbb{R}^L_+$.

Then for any $p \gg 0$, the Hicksian demand correspondence $h(p, u)$ possesses the following properties:

(i) *Homogeneity of degree zero in p:* $h(\alpha p, u) = h(p, u)$ for any p, u and $\alpha > 0$.

(ii) *No excess utility:* For any $x \in h(p, u)$, $u(x) = u$.

(iii) *Convexity/uniqueness:* If $\succsim$ is convex, then $h(p, u)$ is a convex set; and if $\succsim$ is strictly convex, so that $u(\cdot)$ is strictly quasiconcave, then there is a unique element in $h(p, u)$.